

The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane

$\operatorname{Re}(s) > 0.8749570698$

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Abstract

We examine the numerical endpoint in the compensated-probe proof of the seven-eighths zero-free half-plane in [1]. We increase the total prime length and shift the two physical scales in opposite directions. An exact polynomial certificate bounds the resulting high-side exponent. The low comparison includes the same shift in its principal signal. These estimates give the zero-free boundary $4374785349/5000000000 = 0.8749570698$. The accompanying Lean files prove the moment input and continuation argument against the upstream formalization. The full proof passes its axiom audit and Comparator's kernel replay.

1 Statement and geometry

Let $F = \mathbb{Q}(\sqrt{-3})$. As in [1, Equation (1.1b)], let β_* be the supremum of $1/2$ and the real parts of zeros of primitive finite-order Hecke L -functions over F in the critical strip. We use the seven-eighths theorem of [1], so $\beta_* \leq 7/8$. We also use the arithmetic identities, detector estimates, reflected-energy lemma, additive Gram bound, and continuation criterion proved there. The result therefore depends on the correctness of [1]. This note changes only the numerical geometry and the estimates that the change affects. Section 2 checks the changed low comparison. Section 3 supplies the perturbed source moments with zero moment excess, and Section 4 checks the high comparison and analytic domains.

Theorem 1. *Every finite-order Hecke L -function over F and every Dirichlet L -function, including the Riemann zeta function, has no zero in*

$$\operatorname{Re} s > B := \frac{4374785349}{5000000000} = \frac{7}{8} - \frac{214651}{5000000000}.$$

The usual pole at $s = 1$ for a principal character is allowed.

Put

$$\begin{aligned} \epsilon &= \frac{214651}{5000000000}, & \kappa &= \frac{12457523527}{15625000000000}, \\ t &= \frac{42930200831}{250000000000000}, & \ell &= \frac{1}{6} + t, & b &= \frac{1}{8} - 2\kappa, \\ l_x &= \frac{17}{48} + \kappa - \frac{t}{2}, & l_y &= \frac{23}{48} - \kappa - \frac{t}{2}, \\ h &= \frac{13}{16} - \kappa + \frac{3t}{2}, & M &= l_x + l_y = \frac{5}{6} - t. \end{aligned} \tag{1}$$

These lengths satisfy $l_y - l_x = b$, $M + \ell = 1$, and $h = 1 - l_x + \ell$. The general Mellin-exponent identity in [1, Part II, compensated low estimate] gives

$$C(s) = \frac{l_x}{2} + s - 1 + \frac{h}{6} = s - \frac{11}{16} + \frac{\kappa}{3}. \tag{2}$$

Choose finitely many distinct positive slot lengths ℓ_j with $\sum_j \ell_j = \ell$. Their maximum can be made smaller than any prescribed positive mesh. The prime annuli at scales Z^{ℓ_j} are then pairwise disjoint for all sufficiently large Z . The finite physical probe and its exact Poisson identity in [1] make sense with the lengths in (1). All its arithmetic factors are unchanged.

2 The low comparison

Lemma 2. *The normalized compensated probe of [1], formed with (1), satisfies for every $\varepsilon > 0$*

$$|J_\eta(Z)| \ll_{\eta,\varepsilon} Z^{3/16+\kappa/3-t/4+\varepsilon}.$$

Proof. Fix a rescaled subset of slots of total logarithmic length $0 \leq d \leq \ell$. In the notation of the compensated low estimate of [1], its row and surviving-slot lengths are

$$M' = M - 2d, \quad \ell' = \ell - d.$$

The proof of the reflected-row norm in [1, Part II, compensated low estimate] first bounds one branch by M' and the other by $M' + (1 + 3\ell' - 2M')/4$, up to arbitrary small losses. Its general reflected-energy lemma permits these fixed bounded lengths. Substitution gives the squared row norm

$$\sum_{0 < N m \ll Z^{M'}} |B_{m,\sigma}^J(Z)|^2 \ll Z^{M' + ((d-1/6+5t)/4)_+ + \varepsilon}. \quad (3)$$

The additive Gram estimate is available throughout this range. Its scale conditions follow from

$$\begin{aligned} l_x - \ell &= \frac{3}{16} + \kappa - \frac{3t}{2} > 0, & l_y - \ell &= \frac{5}{16} - \kappa - \frac{3t}{2} > 0, \\ M - 2\ell &= \frac{1}{2} - 3t > 0, & l_y - \ell - \frac{11b}{6} &= \frac{1}{12} + \frac{8\kappa}{3} - \frac{3t}{2} > 0. \end{aligned}$$

Thus its P_a^2/Y' term is absorbed by $P_a^{1/6}$, exactly as in [1, Part II, additive Gram bound]. The compensated scale cap is now

$$2l_x - l_y = \frac{11}{48} + 3\kappa - \frac{t}{2}, \quad 2l_x - l_y - \ell = \frac{1}{16} + 3\kappa - \frac{3t}{2} > 0.$$

For a fixed calibration norm q and an unselected tuple norm $L \ll Z^\ell$, these inequalities give, eventually, $qZ^{l_x} \leq Z^{l_y}$ and $LZ^{l_y} \leq q^2 Z^{2l_x}$. They are the physical scale conditions used by the compensated Gram bound. Cauchy-Schwarz, the count of rescaled tuples, and their coefficients add to the basic low exponent

$$-d + \left(\frac{d - 1/6 + 5t}{8} \right)_+.$$

This is nonpositive for $0 \leq d \leq \ell$. Indeed, its positive-part argument is negative until $d > 1/6 - 5t$, and thereafter $d - 1/6 + 5t < 8d$. The empty subset has exponent zero. The direct bound is consequently $Z^{l_x/2+b/12+\varepsilon}$.

The prime normalizer in [1, Part II, principal normalizer] is nonzero and has inverse $Z^{o(1)}$ for every fixed positive slot length. Finally,

$$\frac{l_x}{2} + \frac{b}{12} = \frac{3}{16} + \frac{\kappa}{3} - \frac{t}{4}.$$

This proves the normalized estimate. □

The signal at the proposed boundary is

$$C(B) = \frac{3}{16} + \frac{\kappa}{3} - \epsilon, \quad (4)$$

$$C(B) - \left(\frac{3}{16} + \frac{\kappa}{3} - \frac{t}{4} \right) = \frac{t}{4} - \epsilon = \frac{831}{10^{15}} > 0.$$

The formal continuation assembly uses low loss 10^{-13} , which leaves the strictly positive margin $731/10^{15}$.

3 The perturbed source moments

The fourth-moment lemma with short prime factors in [1] permits moment parameter $\kappa_{\text{mom}} = 3/4$. This parameter is distinct from the geometric skew κ in (1). Its zero-free hypothesis is available because $2\beta_* - 1 \leq 3/4$. The original detector application uses $\kappa_{\text{mom}} = 3/4 + 2\Delta$ with $\Delta = \beta_* - 7/8 > 0$. Here the moment estimates use $\Delta = 0$. The gap $\beta_* - B$, when positive, is used separately to choose the detector losses.

We record the estimates needed for this application. Use the normalized witness polynomials and physical slot factors of [1]. In a source fiber $\mathcal{B}$ at scale $U = Z^d$, write r and m for the inverse and plain witness lengths, $I_{u,n}$ for the inverse witness with n logarithmic derivatives, and $P_{u,j}$ for the plain witness with j logarithmic derivatives. These include the real shifts in $[0, 1]$ and the height shifts in $[-H, H]$. The physical slot factors use $\text{Re } z = 17/50$ and $|\text{Im } z| \leq H$. For a selected set A of slots, let $Q_A(u)$ be their physical product and put

$$\lambda_j = \ell_j/d, \quad L_A = \sum_{j \in A} \lambda_j.$$

Lemma 3. *Suppose $B < \beta_* \leq 7/8$. For every admissible choice of the perturbed detector losses, let $\rho > 0$ be its moment loss. There is a positive mesh μ such that slot lengths with $\ell_j \leq \mu/200$ satisfy the following estimates. After fixing the slots and source, there is an exponent $J \geq 0$ independent of the target character. For each target there are $C_\eta > 0$ and a threshold such that the bounds hold for every source fiber with $1/200 \leq d \leq 7/8$, $51/100 < a \leq 7/8$, and retained rows $Z^{1/100} \leq Nu \leq Z^{d-\rho}$ above that threshold. With $C_H = C_\eta(1+H)^J$, uniformly for $H \geq 0$,*

$$\sum_{u \in \mathcal{B}} |I_{u,n} Q_A(u)|^2 \leq C_H U^{1+\rho}, \quad n \leq 2, \quad r + 2L_A < 1, \quad 2r + 8L_A < 3, \quad (5)$$

$$\sum_{u \in \mathcal{B}} |P_{u,j} P_{u,k} Q_A(u)|^2 \leq C_H U^{1+\rho}, \quad j + k \leq 2, \quad 2m + \frac{9}{2}L_A \leq 1, \quad (6)$$

$$\sum_{u \in \mathcal{B}} |P_{u,j} P_{u,k}|^2 \leq C_H U^{\max\{1, 2m\} + \rho}, \quad j + k \leq 2. \quad (7)$$

The unmarked inverse raw-moment bounds also hold with the count parameters chosen from the fixed source. Together these are the source moment bounds with zero moment excess required by the detector.

Proof. Apply the certified energy theorem of [1] with $\kappa_{\text{mom}} = 3/4$ and stage error $\rho/4$. Its positive-width and zero-slot conclusions give a positive fine mesh μ . The condition $\ell_j \leq \mu/200$ ensures $\lambda_j \leq \mu$ for every $d \geq 1/200$. This does not restrict the total slot length, since the number of distinct positive slots can be increased.

Fix the source's finite excluded prime set S , and let M_S be its product ideal. Use the fixed ideal $M_S \cap (72)$ in the energy source. For each target there are only finitely many

source labels. The norm of the character modulus at every such label is at most $U^{\rho/4}$ once Z is large enough. The exceptional sixth-power-free rows are supported on a fixed finite set of primes and have bounded norm. They are therefore absent from the retained rows for large Z .

The source identities bound the sum of the squared plain witness products by retained energy with the physical prime coefficients. The marked initial state has width $1 + \rho/4$. Its capacity condition follows from $2m + 6\kappa_{\text{mom}}L_A \leq 1$, and the positive-width energy bound gives exponent $1 + \rho/2 \leq 1 + \rho$. For the unmarked pair, use initial width $\max\{1, 2m\} + \rho/4$ and the zero-slot energy bound. This gives (6) and (7).

For the inverse witness, the fixed-source inverse estimates of [1] apply under the two strict capacity inequalities in (5). Distinct slot lengths give disjoint prime supports. Their fixed conductor deletions have modulus-one denominators on these supports for large Z . The inverse raw-moment bound and the marked inverse bound consequently apply to the perturbed slots.

The profile estimates bound all needed seminorms by a fixed power of $1 + H$. Taking a common exponent and summing the finitely many label constants gives J and C_η . The exponent is chosen before the target. Thus the estimates hold at detector height $H = Z^{2\tau}$, and the usual late choice of $\tau > 0$ absorbs the polynomial height factor into the prescribed detector loss. $\square$

4 The high comparison

We first record the extension of the Euler regions used by the contour arguments. The local factors below are those of [1, Equations (5.13a)–(5.13c)].

Lemma 4. *The full compensated correction is holomorphic, with the bounds needed for principal-row contour moves, on the second Euler region of [1] with its condition $\text{Re } s \geq 7/8$ replaced by $\text{Re } s \geq B$. The principal local quotient there satisfies*

$$G_p/H_p = -1 + O((Np)^{-B}).$$

The small-row contour can use the first Euler region with $\epsilon_0 = 37/100$.

Proof. The exact local defect formula in [1, local Euler lemma] bounds $H_p - 1$ in the original second region by $O(Q^{-363/200})$ for $p \nmid u$ and $O(Q^{-33/40})$ for $p \mid u$, where $Q = Np$. Every local exponent involving $\text{Re } s$ has coefficient of absolute value at most six. Moving its lower boundary left by ϵ therefore gives the respective bounds

$$O(Q^{-363/200+6\epsilon}) \quad \text{and} \quad O(Q^{-33/40+6\epsilon}).$$

The first exponent is less than -1 . The unramified product converges normally, while the ramified product has the same $O_\epsilon((Nu)^\epsilon)$ bound. For $u = 1$ its tail still tends to zero faster than $P_0^{-4/5}$. Enlarging the fixed excluded set therefore gives the same uniform nonvanishing of the principal product.

The four errors for the principal selected factor have exponents $-\text{Re } s$, $-6\text{Re } z$, $4 - 5\text{Re } s - 6\text{Re } z$, and $1 - \text{Re } w - 6\text{Re } z$. On the extended region they are all at most $-B$. The formula for G_p has no division by P_p or H_p , so the full finite selected-tuple correction remains holomorphic at their zeros. The bounds above and finite slot count give the all-height majorant used in the principal contour proof of [1].

For small rows, $\beta_* > B$ implies $\beta_* + e + 1/2 > 1 + 37/100$. The first Euler region of [1] allows any positive ϵ_0 , so its unchanged local estimates apply to the line $(\beta_* + e, 1/2, 17/50)$. The central bin contours already lie in its region with $\epsilon_0 = 10e$. These are the only analytic

domain uses of the original $7/8$ lower bound in the shared contour lemmas. Repeating their proofs with the extended region gives the same principal, bin, and outer-row contour estimates. $\square$

Lemma 5. *If $B < \beta_* \leq 7/8$, then for some $\sigma > 0$ independent of the target character,*

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*) - \sigma},$$

where f_η is the principal Mellin signal from [1] with (2).

Proof. Lemma 3 supplies the fourth-moment estimate at $\kappa_{\text{mom}} = 3/4$, because its zero-free input is $\beta_* \leq (1 + \kappa_{\text{mom}})/2 = 7/8$. Every detected bin has $\delta = 2a - 1 \leq 2\beta_* - 1 \leq 3/4$. Its plain prime capacity is thus the baseline capacity in the proof of [1, Part II, capacity comparison], with no positive $\Delta/4$ surcharge. The row exponent is bounded by $R_*(\delta, x)$ plus an arbitrarily small loss, with $0 \leq x \leq 1/2$ and $q = x\delta$. The same selected-prime argument applies since $\ell/h > 8/39 > 7/37$ and $5\ell - h > 1/48$. A fixed sufficiently fine even slot mesh meets all rounding conditions.

At row length $d = h$, the general high exponent relative to $C(B)$ is

$$E_B(h) = a - B + h \left(\frac{17}{50} - \frac{1}{6} \right) - a\ell_y - \frac{\ell}{2} + q\ell + h \left(R + \frac{\delta}{2} - \frac{17}{50} \right). \quad (8)$$

Using $\delta = 2a - 1$ and holding a, δ, q, R fixed while changing (1) gives

$$E_B(h) - E_{7/8}(13/16) = \epsilon + t \left(-\frac{1}{2} + \delta + q + \frac{3R}{2} \right) + \kappa \left(\frac{2}{3} - R \right). \quad (9)$$

We evaluate the combined exponent using the exact balanced row count from [1, Part II, compensated endpoint certificate]. Put $y = 1/2 - x$ and define

$$D(y) = \frac{37 + 34y}{18}, \quad P(y) = \frac{7 + 18y + 8y^2}{9}, \quad J(\delta, y) = (5/6 - \delta)D(y) + \delta P(y).$$

Then $J \geq 35/54 > 0$ on the parameter box, and

$$R_* = 1 - \delta + \frac{(5/6 - \delta)\delta P(y)}{2J(\delta, y)}.$$

With $R = R_*$ and the value of t in (1), direct substitution gives

$$16200000000000000000J(\delta, y) \left(-E_B(h) - \frac{1}{1250000000000} \right) = H(\delta, y),$$

$$H(\delta, y) := A(y)\delta^2 + L(y)\delta + K(y),$$

where

$$\begin{aligned} A(y) &= 38381036152738950 + 98112671277888624y \\ &\quad + 70349424015981576y^2 + 24024727795678656y^3, \\ L(y) &= -29828491018092129 - 47206874749477924y \\ &\quad - 3166469409426772y^2, \\ K(y) &= 5795432885635430 + 5325532921935260y. \end{aligned}$$

The following identity certifies $H \geq 0$ for every $y \geq 0$:

$$\begin{aligned} 4A(y)H(\delta, y) = & (2A(y)\delta + L(y))^2 + 979807389619512241359 \\ & + 275799814009247843319007787580888y \\ & + 1303439473219357232720782491774728y^2 \\ & + 1756569235069508453488622075444704y^3 \\ & + 501751386744843713686464752902256y^4. \end{aligned}$$

Every coefficient in the remainder is positive, as is $A(y)$. Hence

$$-E_B(h) \geq \frac{1}{1250000000000}. \quad (10)$$

For a row-count loss $0 \leq \rho \leq 1/32$, put $R = R_* + \rho$. The source bounds imply $0 \leq R \leq 3/2$ and $0 \leq R + \delta/2 - 17/50 \leq 2$. For $\zeta, \mu \geq 0$, frequency length $v \leq h + \zeta$, and row scale d with $d - v \leq \mu$, the mixed exponent differs from its balanced endpoint by

$$h\rho + (v - h)(R + \delta/2 - 17/50) + (d - v)R \leq h\rho + 2\zeta + \frac{3}{2}\mu.$$

These losses, the additional physical-scale cost $t(3e + \text{mesh}) + 6\kappa e$, and the remaining detector losses can all be chosen below the margin in (10).

The floor and intermediate-row margins in [1] are respectively $7/1200$ and $49/14400$ at the original geometry. The floor shift costs $t(121/40 + 3e + \text{mesh}) + \kappa(67/100 + 6e)$, including the changed row threshold. The parameter budget absorbs this cost and the boundary change $7/8 - B = \epsilon$. In particular,

$$\epsilon + \frac{121t}{40} + \frac{67\kappa}{100} < \min\left(\frac{7}{1200}, \frac{49}{14400}\right).$$

The intermediate estimate can also be checked directly. Before the small detector losses, its bounds are $d \leq 1/2$, $q \leq \delta/2$, and $R \leq 76/75 - 2\delta/3$. Its exponent relative to $C(B)$ is therefore at most

$$-\frac{529}{2400} + \frac{25\delta}{96} + \epsilon + t\left(\frac{1}{100} + \frac{3\delta}{4}\right) + \kappa\left(\frac{49}{150} + \frac{\delta}{2}\right) < -\frac{3}{125} \quad (0 \leq \delta \leq 3/4).$$

The expression increases with δ , and substitution of $\delta = 3/4$ gives $-2440350045024453/10^{17}$. The small-row margin is $63/800$. Relative to the shifted signal, its geometric cost is

$$\frac{51t}{100} + \frac{49\kappa}{150} < \frac{63}{800}.$$

For large rows, the absolute contour can still be taken sufficiently far right after choosing ζ . Lemma 4 supplies the principal contour, whose margins $l_y/20$ and $h/600$ remain positive. Its local approximation supplies a positive power saving in each fixed slot. The prime normalizer costs only $Z^{o(1)}$.

All detector, mesh, real-contour, and normalizer losses can be chosen below these fixed positive margins before the target character is fixed. The explicit central budget takes a common small parameter at most 10^{-16} and bounds the later detector allowances by $\text{small}/(N + 2000)$, where N is the fixed slot count. It reserves the high margin in (10), including the row-count, physical-scale, Mellin, and normalization costs. The late height choice in [1] then gives one $\sigma > 0$ for every target, as claimed. $\square$

Proof of Theorem 1. Suppose $\beta_* > B$. Lemma 4 supplies a principal factor holomorphic and bounded away from zero on $\operatorname{Re} s > B$. Lemma 2 and (4) give the direct bound required by the continuation criterion of [1, Section 2]. Its parameter ω can be any sufficiently small positive number below $\beta_* - B$. Lemma 5 gives the other required bound with a target-independent saving. The criterion contradicts $\beta_* > B$. Hence $\beta_* \leq B$. The transfer in [1] from primitive Hecke characters to imprimitive Hecke and Dirichlet characters proves the stated assertions. $\square$

Verification scope

The rational inequalities in (4) and (10) can be checked independently by the accompanying `scripts/verify_bounds.py`. The accompanying files in `lean/` contain proofs of the geometry, reflected-exponent calculation, high-side endpoint, normalized compensated low estimate, local Euler bounds, and principal and global Euler product extensions against the upstream OAI formalization. The finite source multiplier bounds and the principal source's w and z contour shifts, ordered double shift, uniform contour tails, and x contour transport also have Lean proofs at the new boundary. The selected high-row local factors, weighted prime-slot sum, physical-row dyad, Mellin integral, and changed small- and large-row tail bounds, central cube truncation error, canonical ray-cube choice, and normalized floor-row collection have proof files there as well. The principal slot-ratio and normalized residue proofs use the new boundary. The physical principal remainder at the changed lengths is bounded on a fixed contour just right of $7/8$, leaving a power saving of $1/4000$ relative to β_* in the proposed interval. The contour at $\beta_* + e$ used in the final comparison retains a saving of $1/3000$. The perturbed source moment transport is implemented in `OAIHighMomentTransport.lean`. It supplies the chosen fourth-moment input used by the central-bin estimate and detector assembly. `OAIHighUnconditional.lean` derives the prior $\beta_* \leq 7/8$ bound from the upstream unconditional module and concludes the stronger Hecke, Dirichlet, and zeta nonvanishing statements without a moment hypothesis. The final theorem's axiom audit passes with only the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. It rejects any other transitive axiom dependency, including incomplete proofs. The verification script requires the pinned upstream commit and a clean upstream working tree. Full Lean verification and Comparator replay of the updated bound pass. The Comparator challenge uses upstream's Hecke definitions with the bound changed to B . Comparator checks all three nonvanishing statements and replays their proofs through the Lean kernel. The proof uses Lean 4.34.1 and upstream revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Compiling this note checks the TeX. The separate Lean build checks the proof. The stronger boundary in Theorem 1 is an extension of [1]'s stated theorem.

This note, its Lean proofs, and the verification scripts were entirely AI generated.

References

- [1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\operatorname{Re}(s) > 7/8$* , 30 September 2026. Source consulted 7 October 2026. [Source TeX at the verified revision](#).